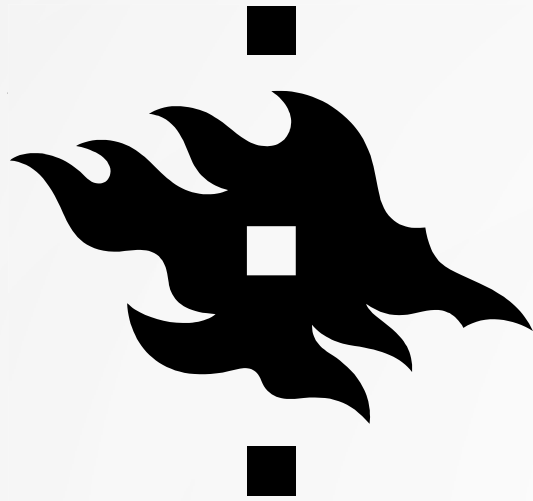




UTILIZATION OF LOW-COST SENSORS ASSISTED WITH MACHINE LEARNING IN MONITORING THE PERSONAL EXPOSURE

Tareq Hussein

How to quantify air pollution exposure levels?!

Experimental methods:

- State-of-the-art setup: research-grade instruments provide detailed information about the physical and chemical characteristics of aerosols and the environment
- Monitoring Stations: dedicated for monitoring air quality indicators and usually they operate as a network
- High-cost instruments: research-grade instruments not at the level of the state-of-the-art, but these are accurate and reliable
- Medium-cost instruments: limited accuracy

The data-base produced **lacks good spatial-resolution**, but have good **high time-resolution** and **extensive information**

Low-cost sensors:

- **easy** to deploy and handle
- Operated indoors and outdoors as well as on moving platforms (**carry on/portable**)
- Suffer of drifting -- require **frequent calibration**
- lack of accuracy – improved by **machine learning (ML)/artificial intelligence (AI)**
- High **spatio-temporal resolution**
- Can be upgraded as **virtual sensors**
- Deployed as subnetwork under the monitoring station network

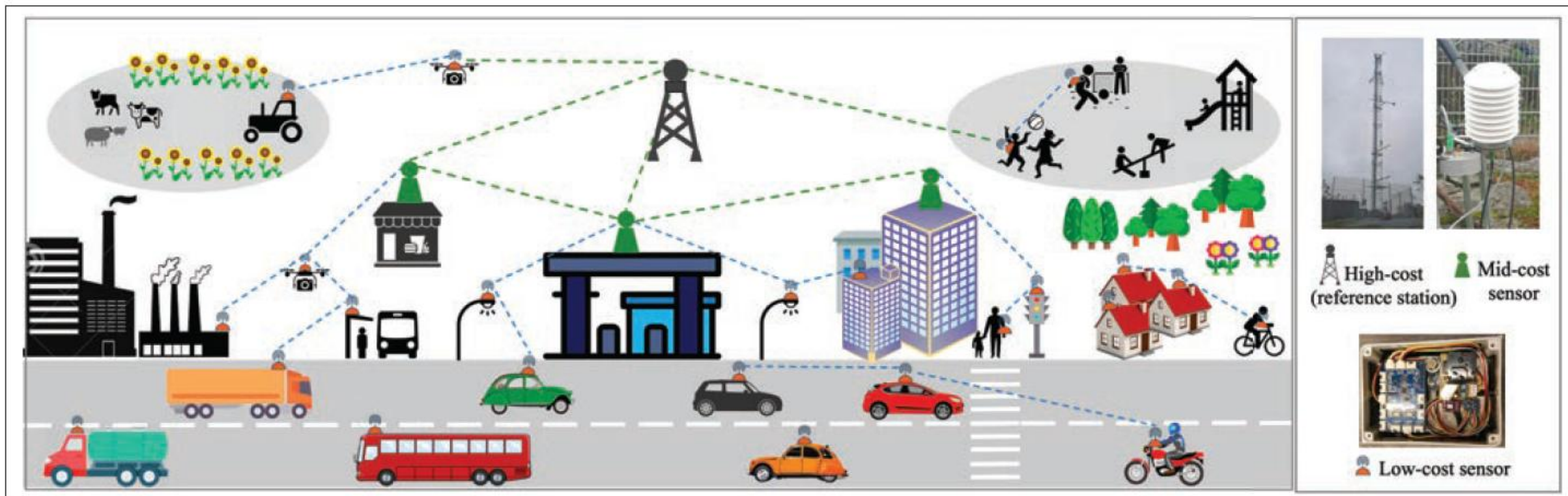


Figure 1. Vision of massive scale air quality monitoring with opportunistic sensor calibration. Green lines denote calibration links between reference and mid-cost sensors, and blue lines denote calibration links between low-cost sensors.

Model simulation and prediction:

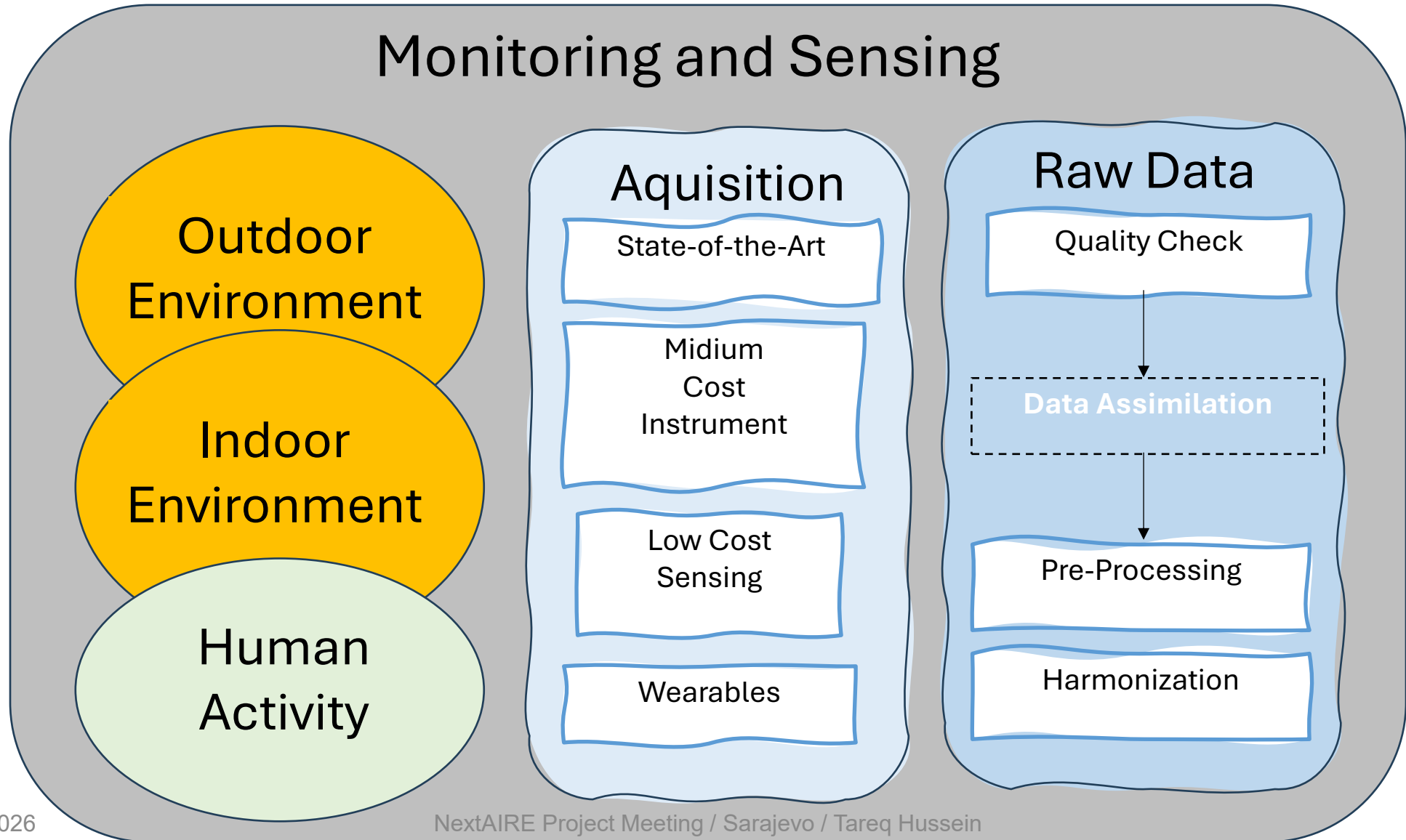
- Indoor Aerosol Modelling
- Urban Aerosol Modelling
- Statistical predictive models
- Machine learning (ML) and artificial intelligence (AI)
- Proxy concept

Strength:

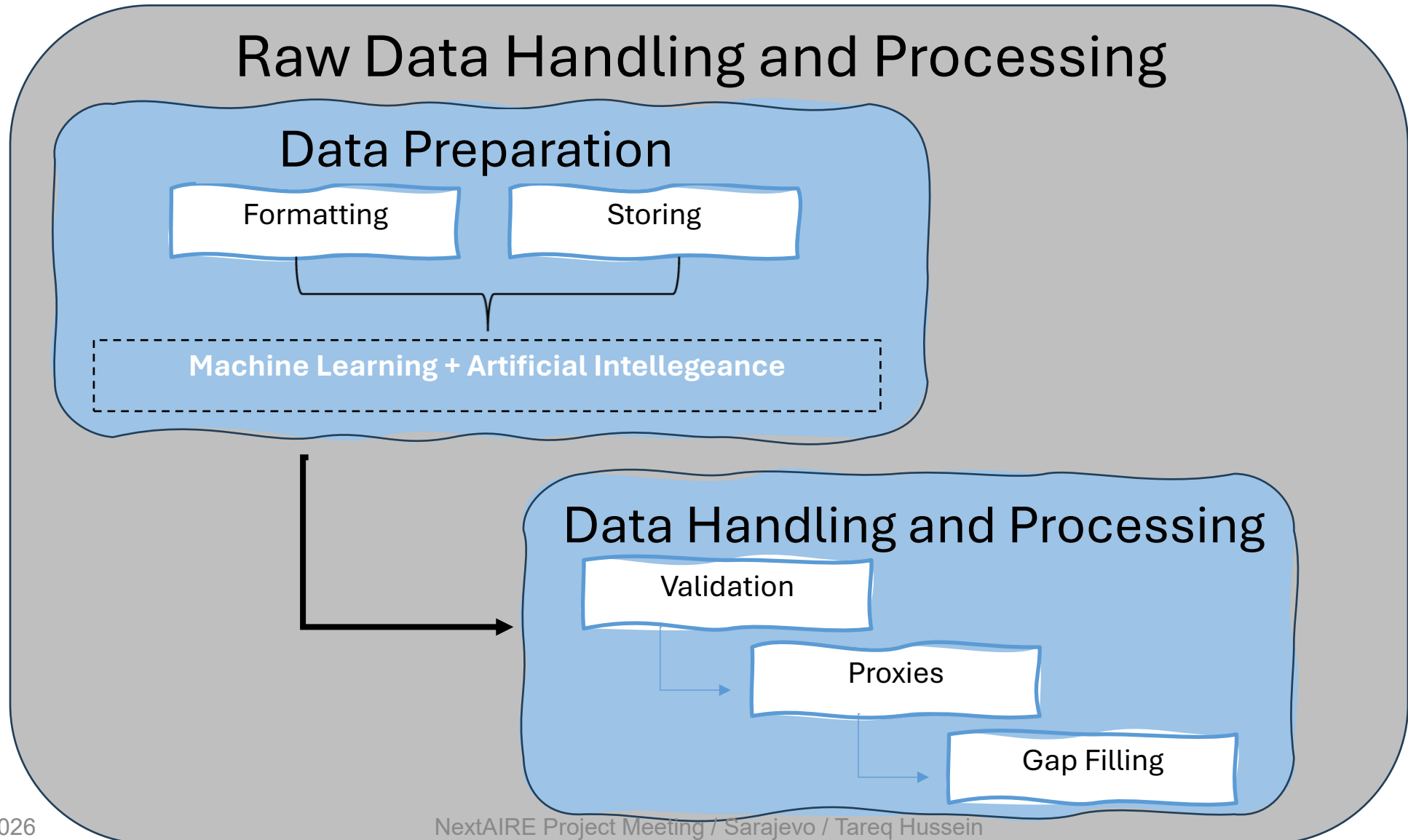
- Fill in gaps in data
- Forward/backward in time
- Boost the data-base beyond the measurement

FRAMEWORK

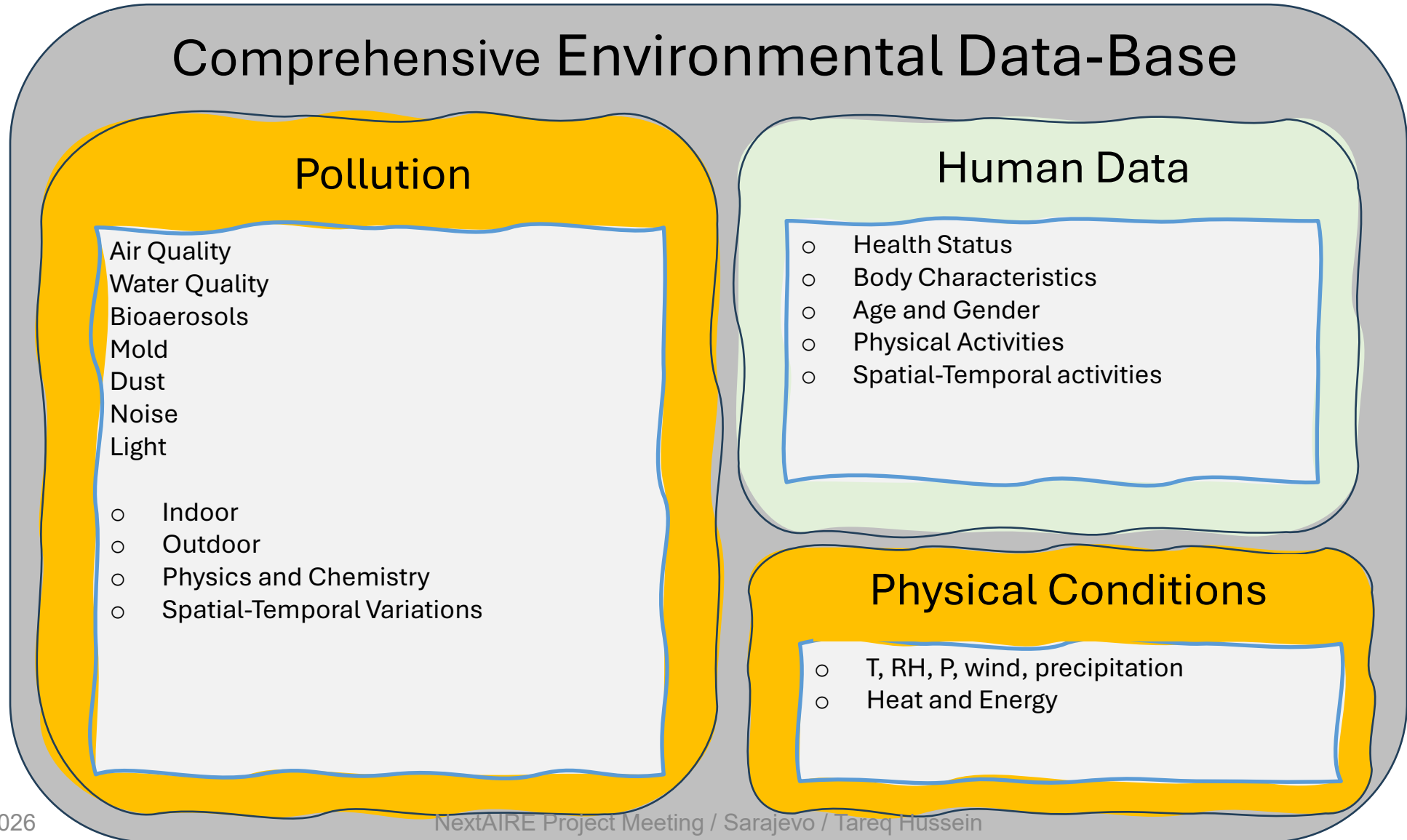
Framework – Phase I



Framework – Phase II



Framework – Phase III



Development and Utilization of Proxies

BOOSTING SENSOR CAPABILITIES

Low-Cost Sensor Boosting Capabilities – Gap in setup

A LCS sensor **can** monitor (physically) some air quality indicators including (not all on the same device!!):

- Particulate Matter: $PM_{2.5}$, PM_{10} , etc.

- Particle Number Size Distribution: 0.3 (0.1) μm – 10 (25) μm

- Gases: CO_2 , NO_x , O_3 , etc.

- Environmental: T, RH, and P

A LCS sensor **can't** monitor (physically) some air quality indicators including (not all on the same device!!):

- Particulate Matter: $PM_{0.1}$

- Particle Number Size Distribution: $D_p < 0.3$ (0.1) μm

- Lung Deposition Surface Area

- Black Carbon

- Gases: CO, NO_2 , NO, etc.

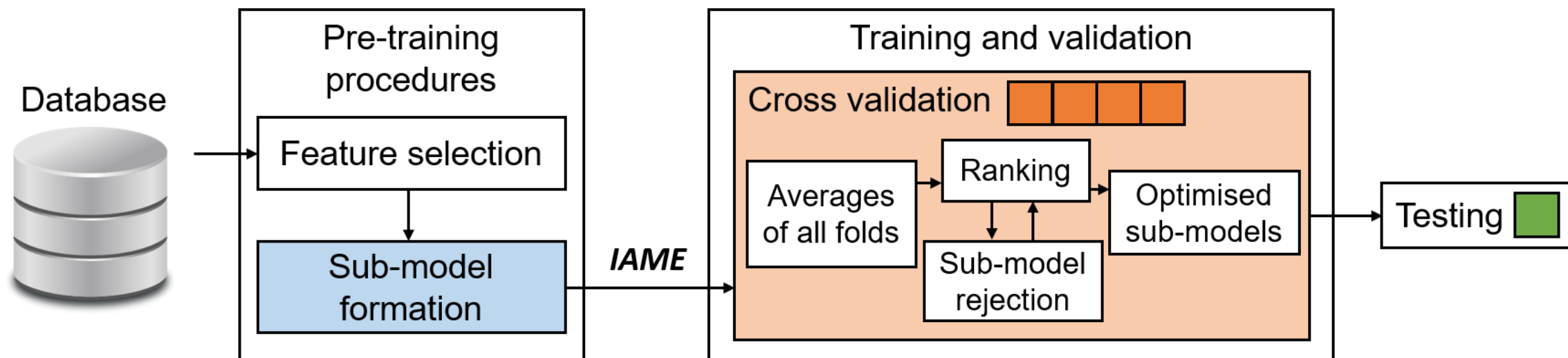
Low-Cost Sensor Boosting Capabilities – Proxy

The **Proxy** is a mathematical estimation for a variable based on the knowledge of other variable(s):

Step 1: how the estimated variable is related to the estimating variable(s)

Step 2: validation and testing

Step 3: estimation



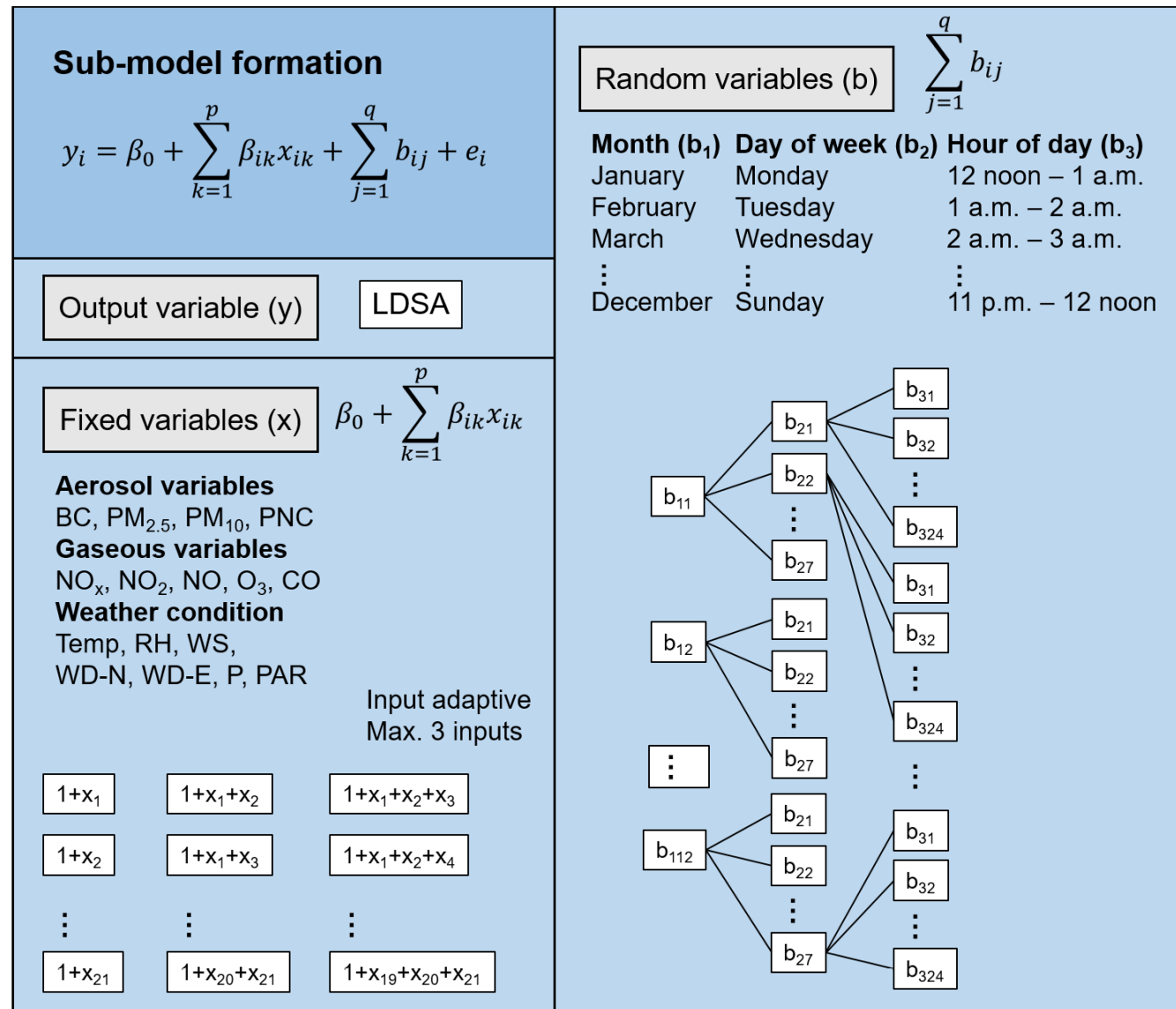
Low-Cost Sensor Boosting Capabilities – Proxy

The **Proxy** can be developed and assessed by numerical and statistical models/analysis.

A group of variables can be cross-learned so that the proxy can work in different direction across the data-base variables.

Benefit:

- Gap filling
- Backward/forward prediction
- Understanding processes

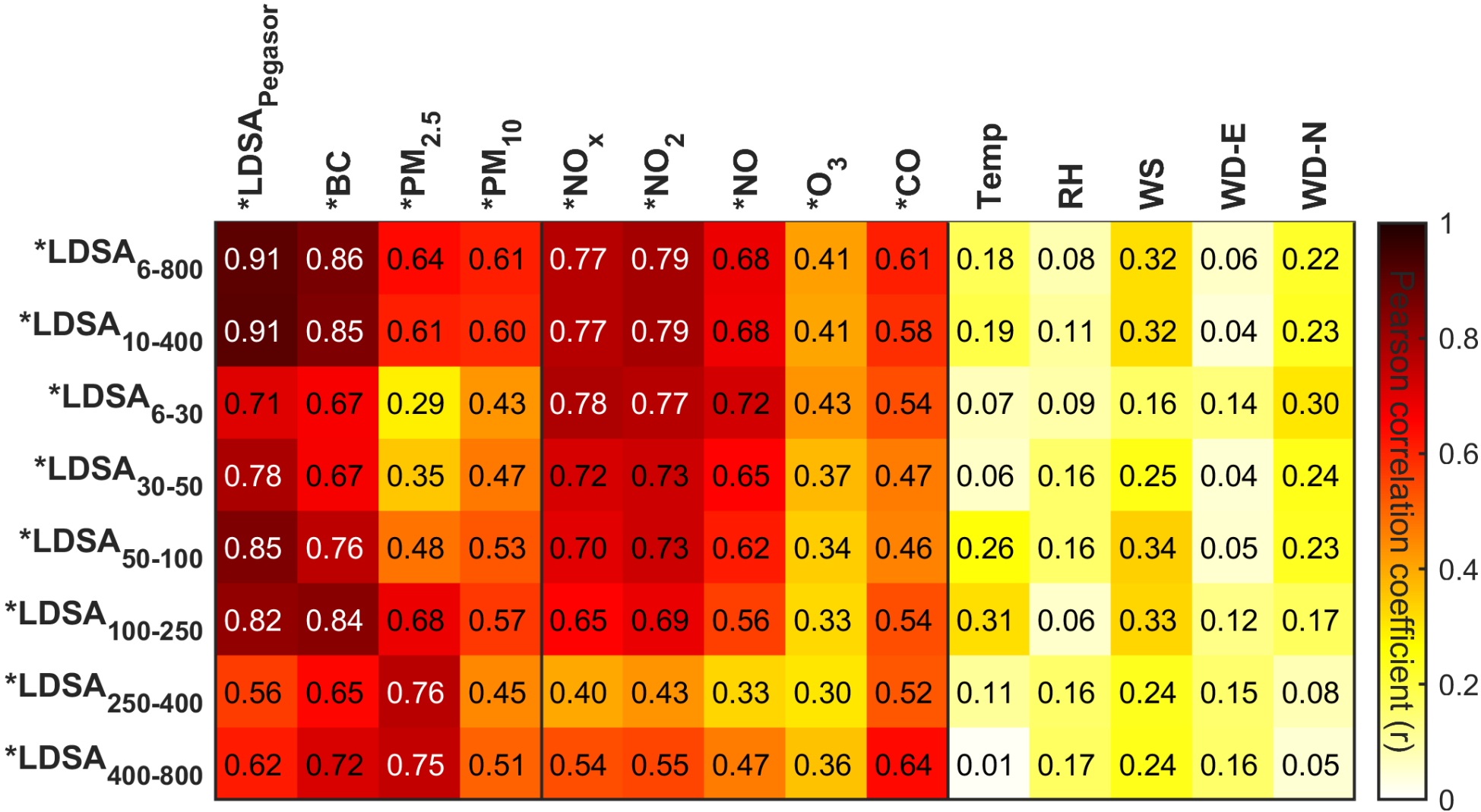


Example I – LDSA Proxy

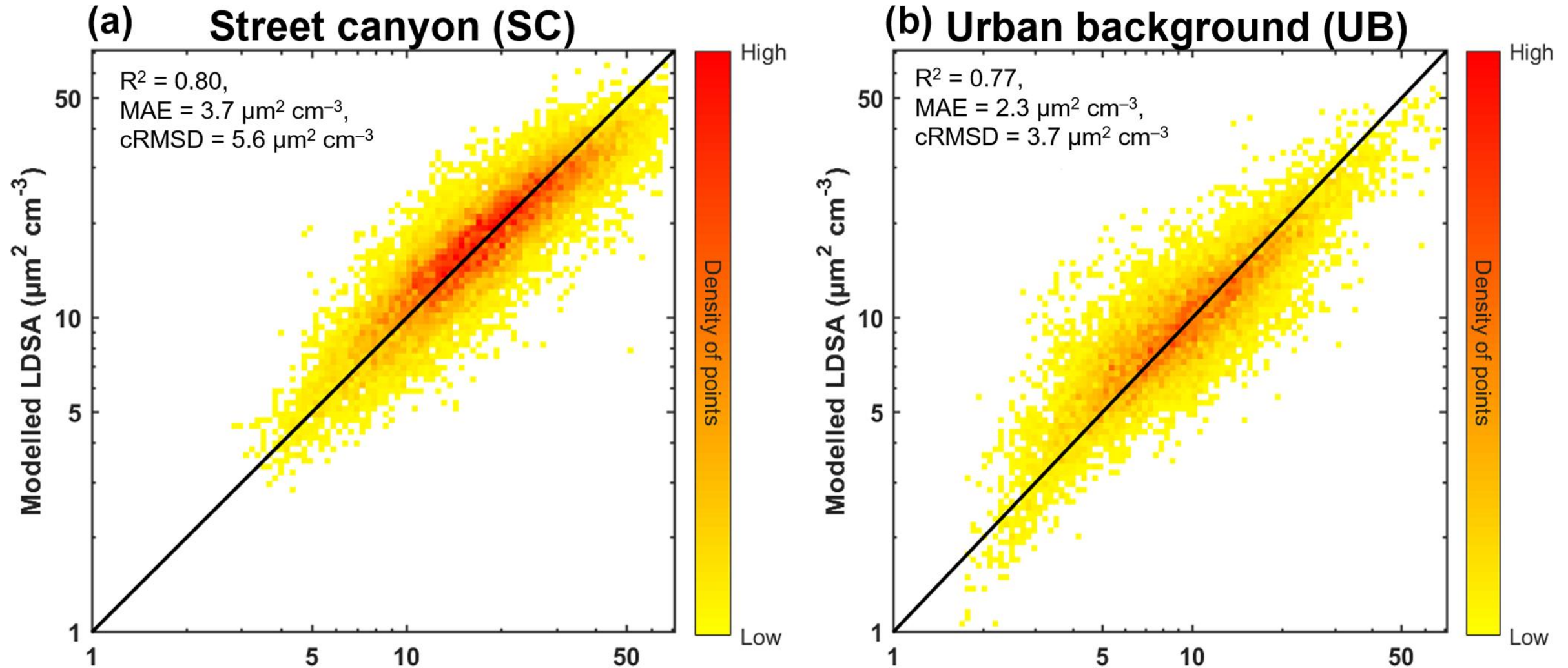
LUNG DEPOSITION SURFACE AREA

Example! Lung Deposition Surface Area (LDSA) Proxy

Pearson correlation coefficient heatmap of LDSA of different particle size fractions (in nm) by ICRP lung deposition model and the other air quality indicators (Helsinki street canyon).



Example! Lung Deposition Surface Area (LDSA) Proxy



Example II – BC Proxy

BLACK CARBON

Example! Black Carbon (BC) Proxy

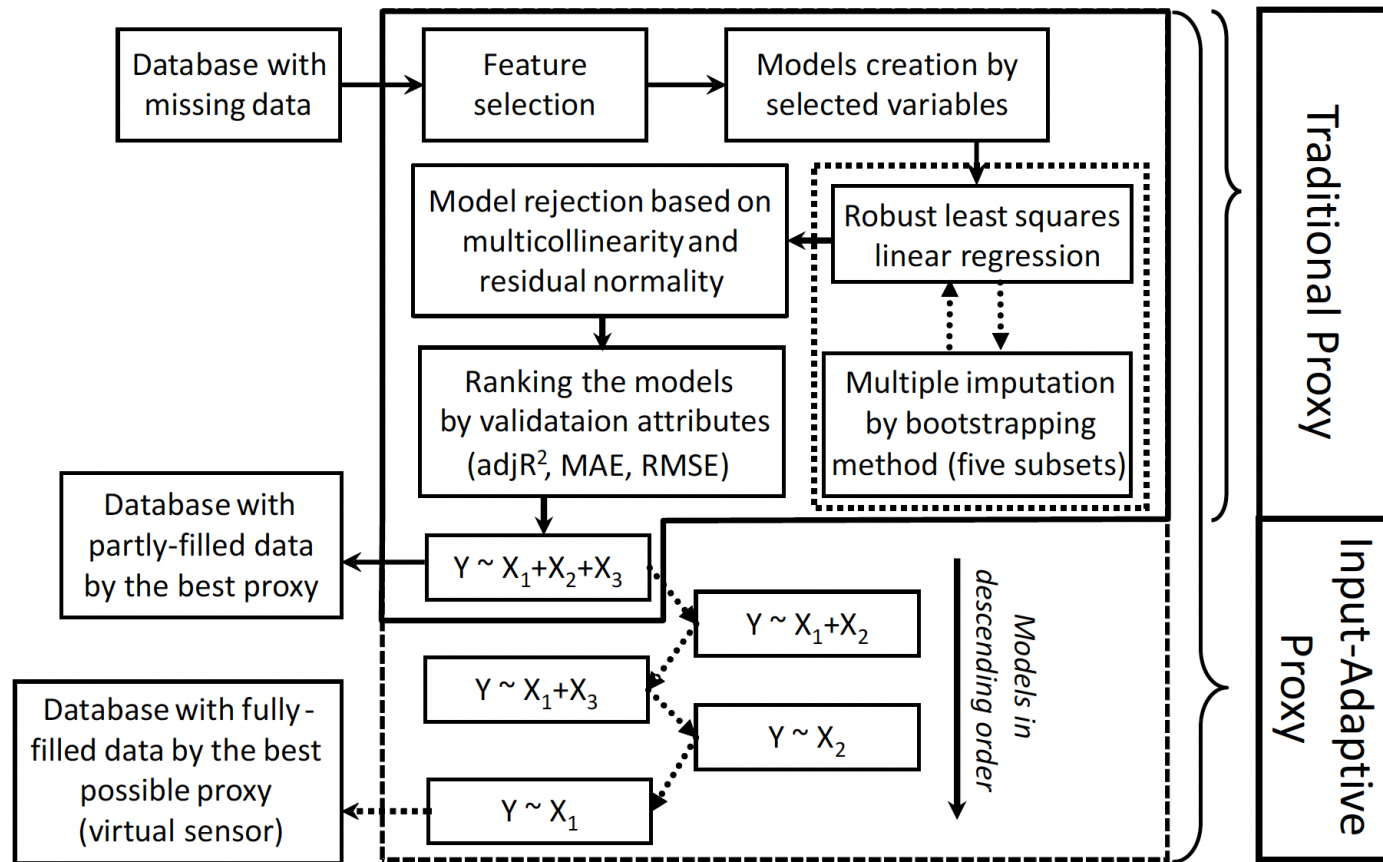
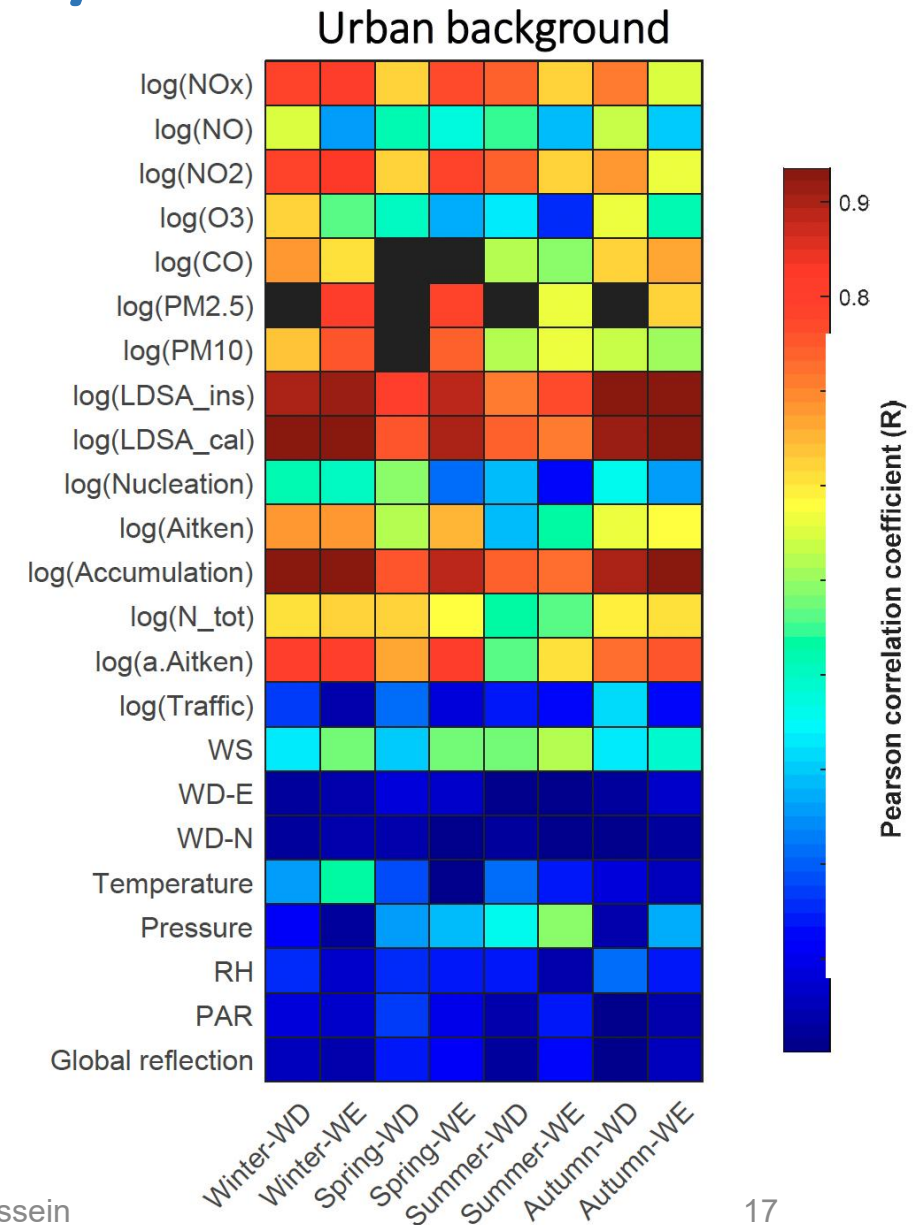
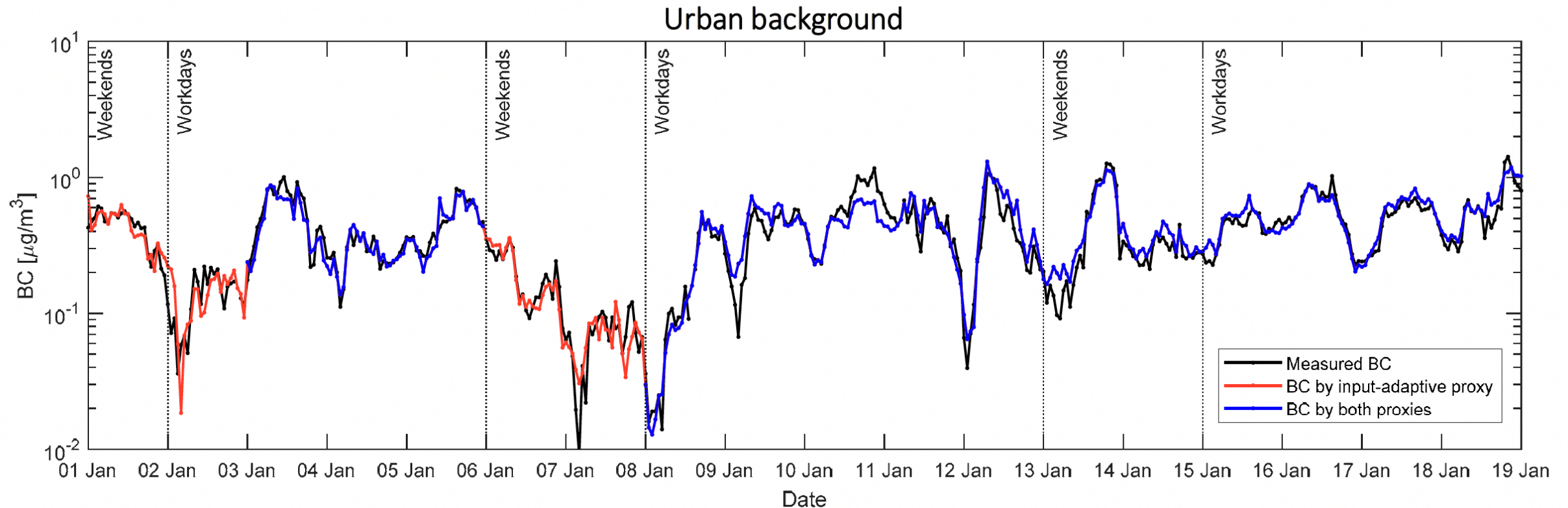


Figure 1. Schematic diagram of the procedures of developing the recursive input-adaptive proxy.



Example! Black Carbon (BC) Proxy



Example! Black Carbon (BC) Simple Proxy

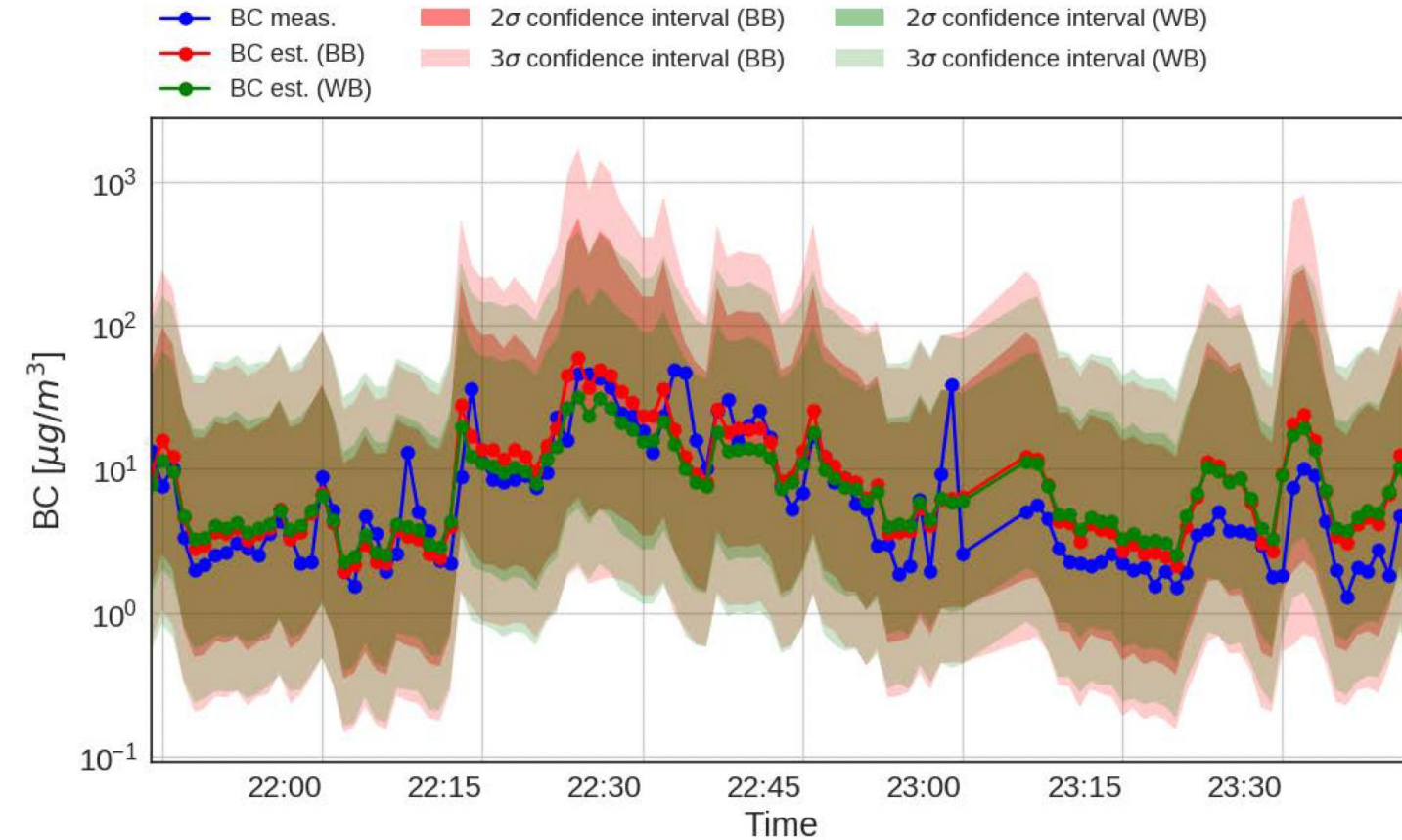


Figure 4. Time-series data of BC mass concentrations in the Amman city center. The blue dot is the measured BC, whereas the red dashed line and the light green region are the measured BC and its 2σ uncertainty.

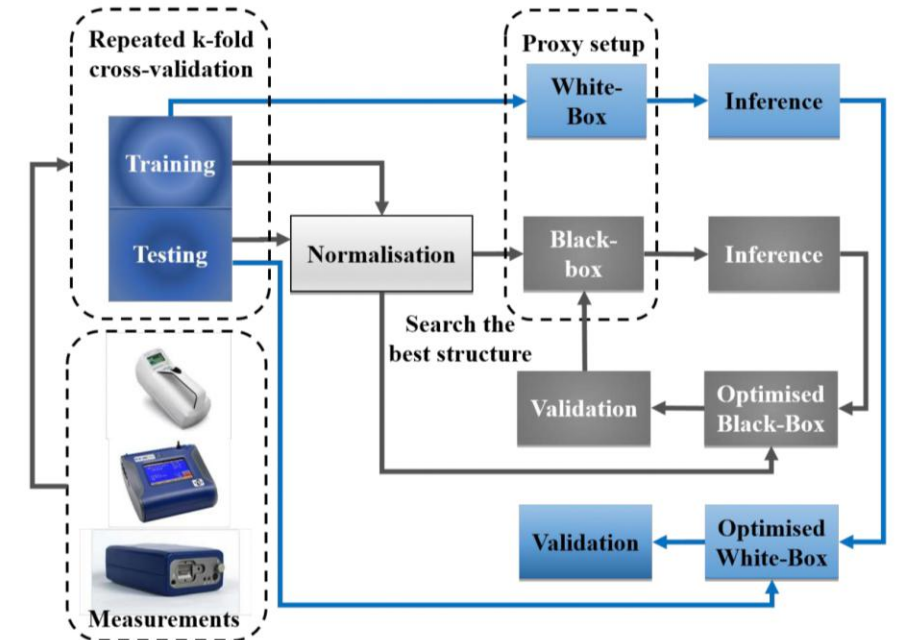


Figure 3. The procedure of the modelling processes for the white-box (WB) and black-box (BB) proxies.

Example III

UTILIZATION OUTDOORS

Utilization of LCS in Amman, Jordan

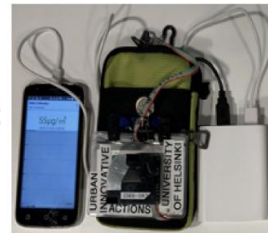
- One year
- Deployment of five (5) LCS
- In-field calibration against:
 - Seven (7) monitoring stations
 - Two (2) reference instruments
- PM_{2.5}

DustTrak
DRX 8534



PM_{2.5}

Panasonic
(GA1)



PM_{2.5}

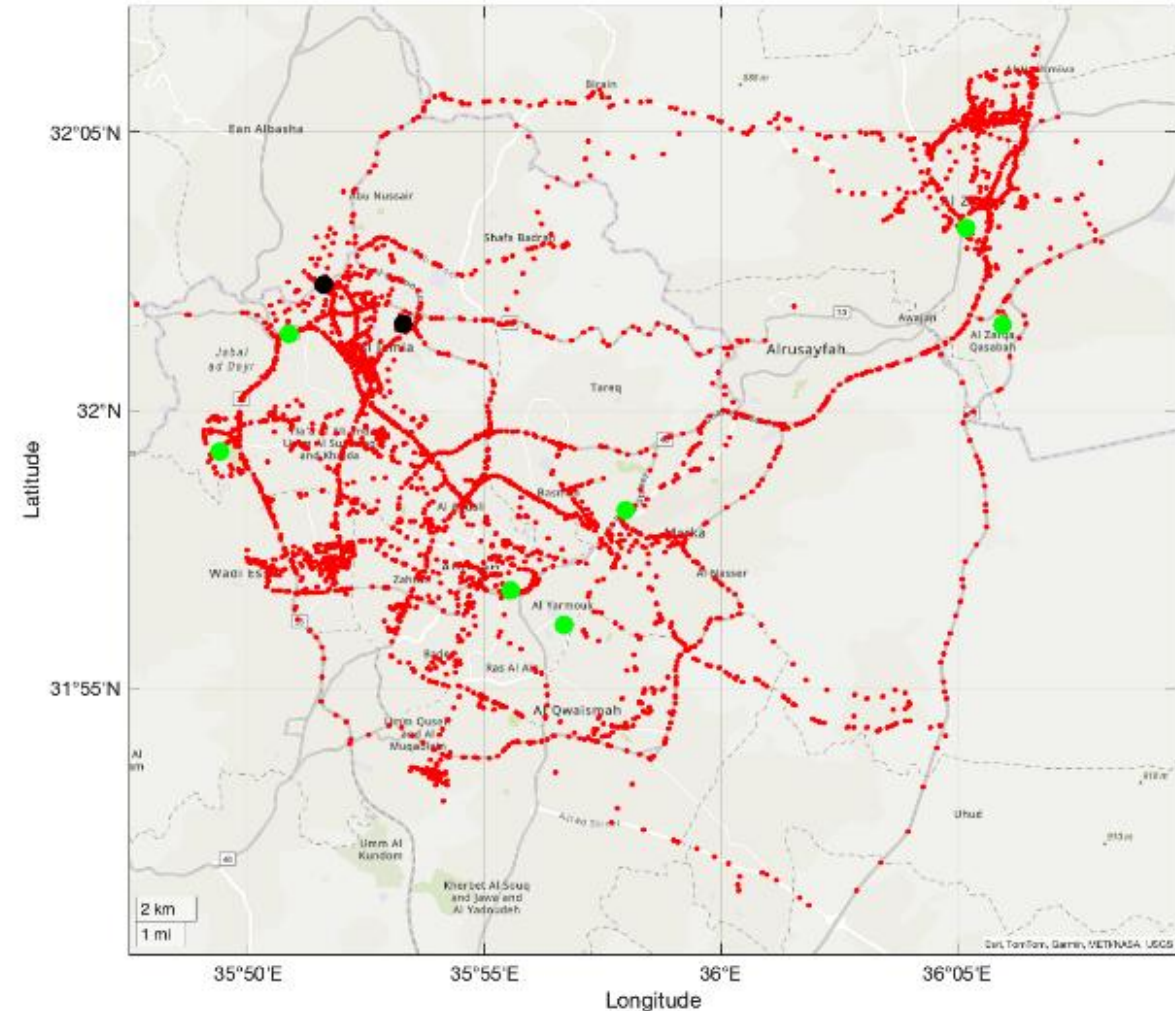
Met.
Var.

Legend:

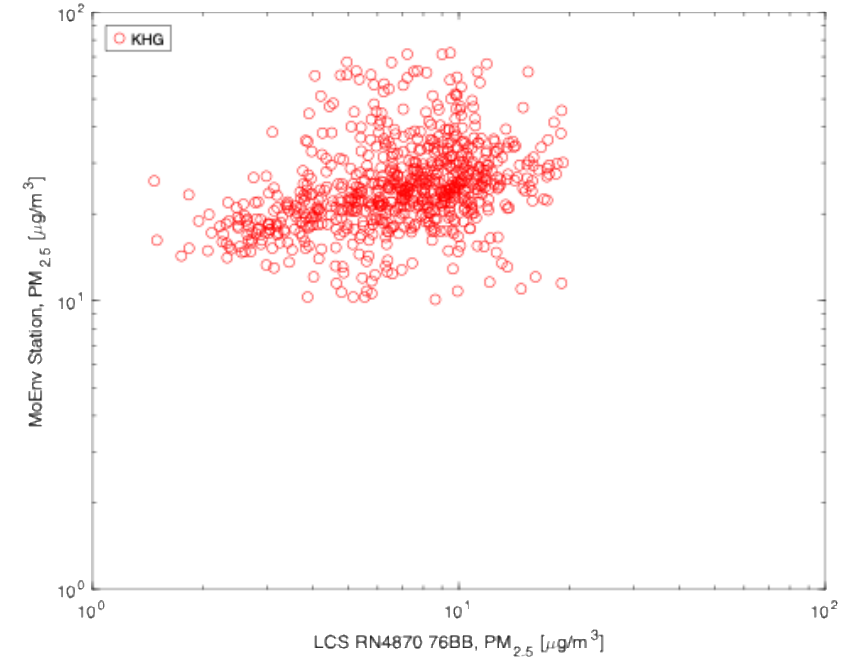
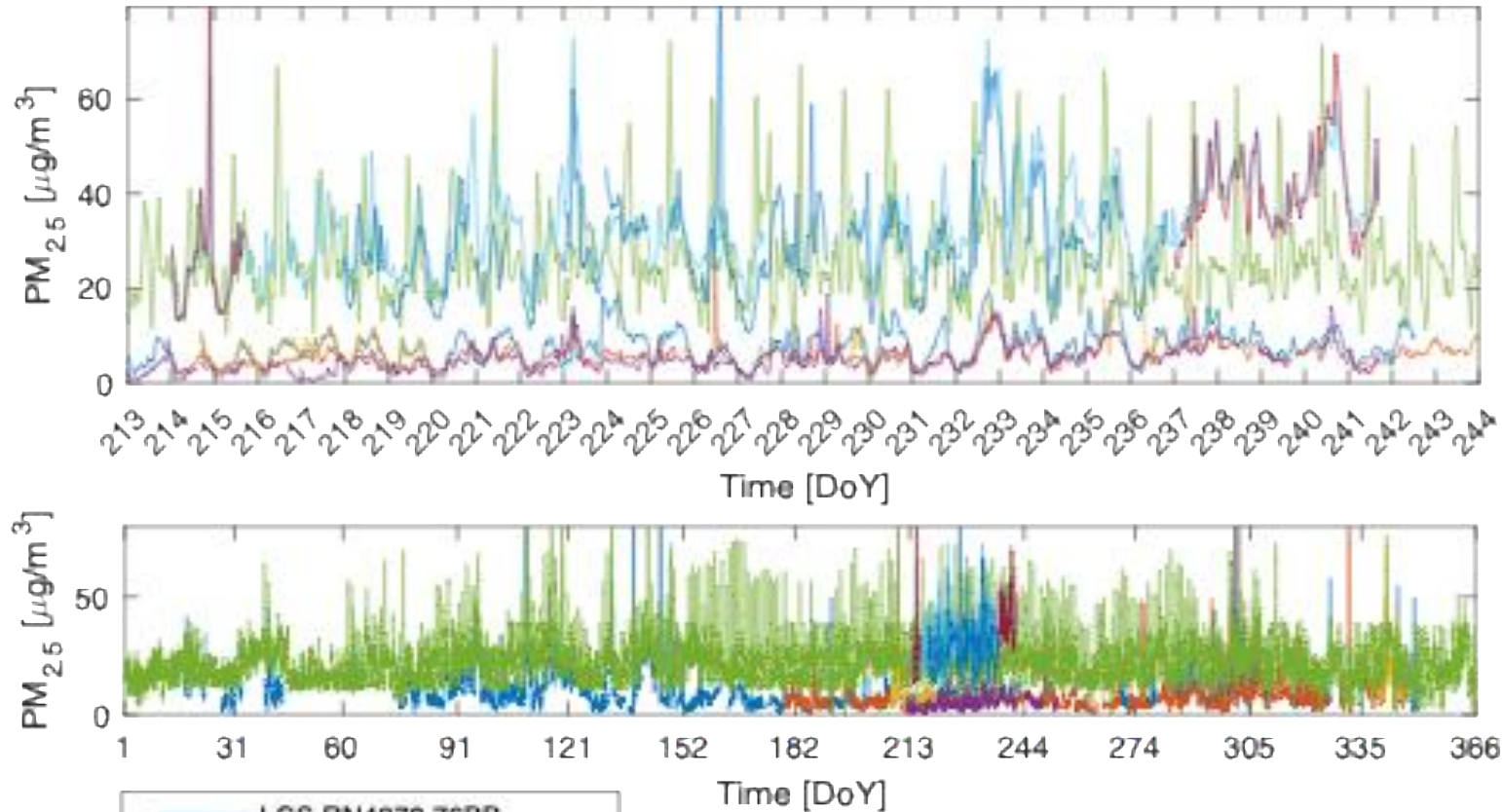
Red == mobile LCS

Green == station

Black == reference instrument



Utilization of LCS in Amman, Jordan



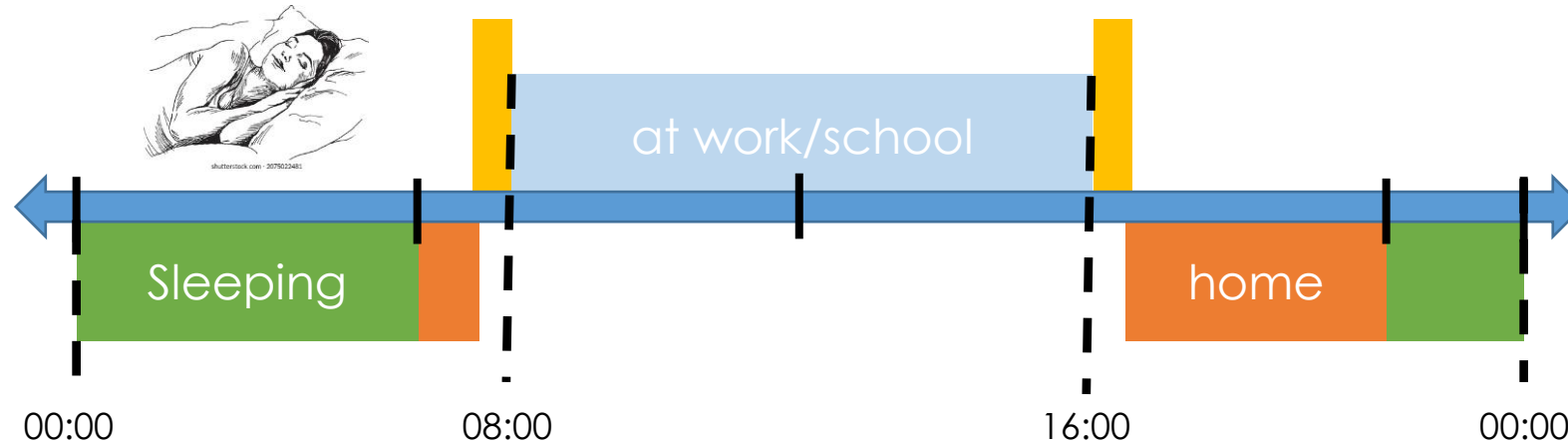
- The LCS underestimates LCS monitoring
- Require in-field calibration

HUMAN ACTIVITIES AND BEHAVIOR

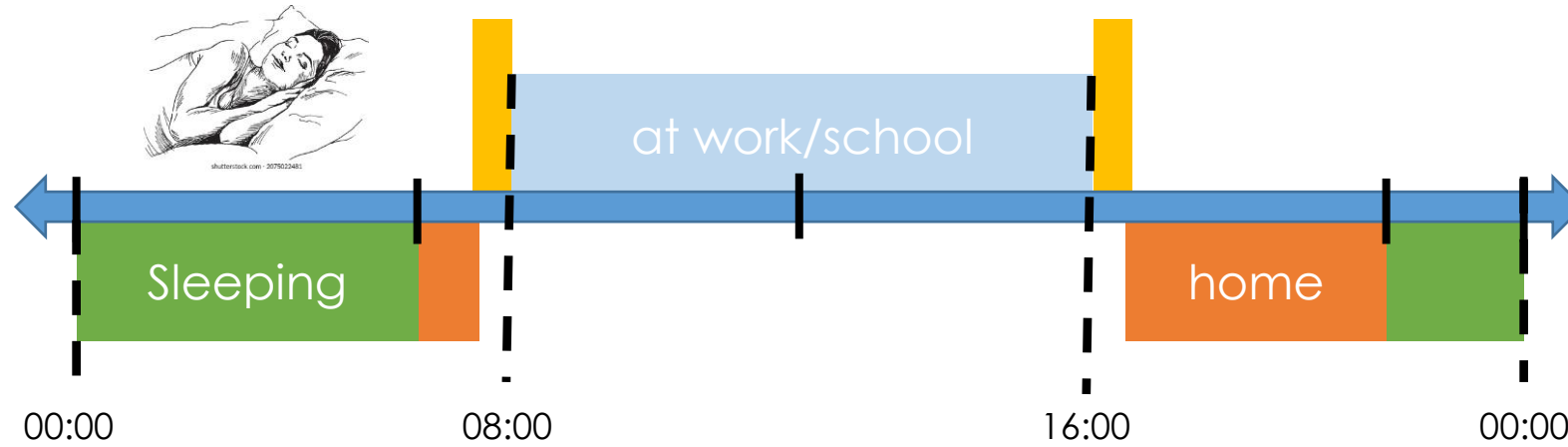
Gender and Age



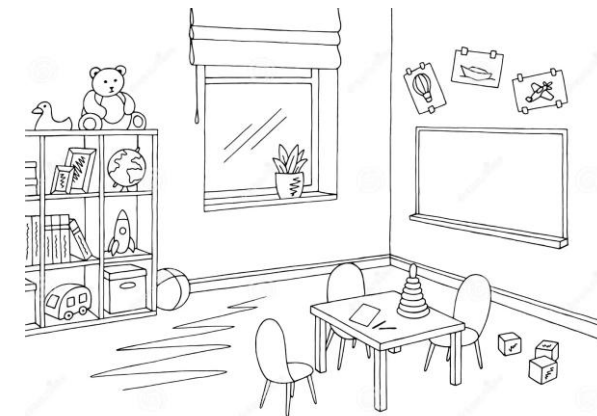
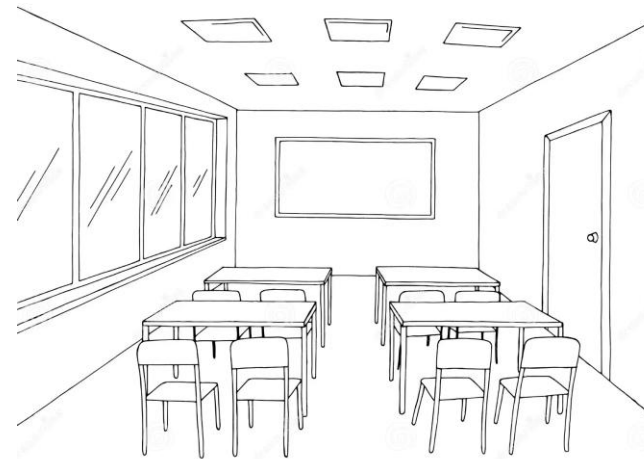
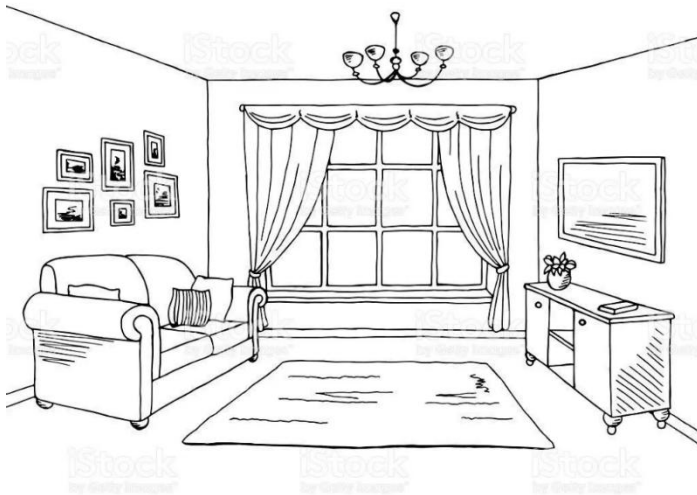
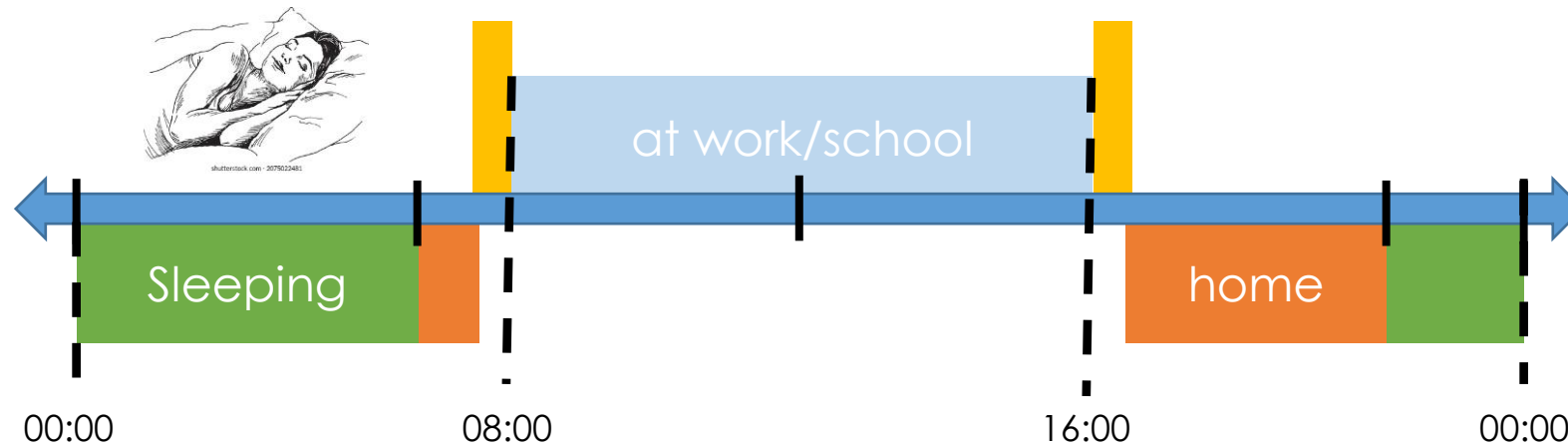
Daily Routine



Activity



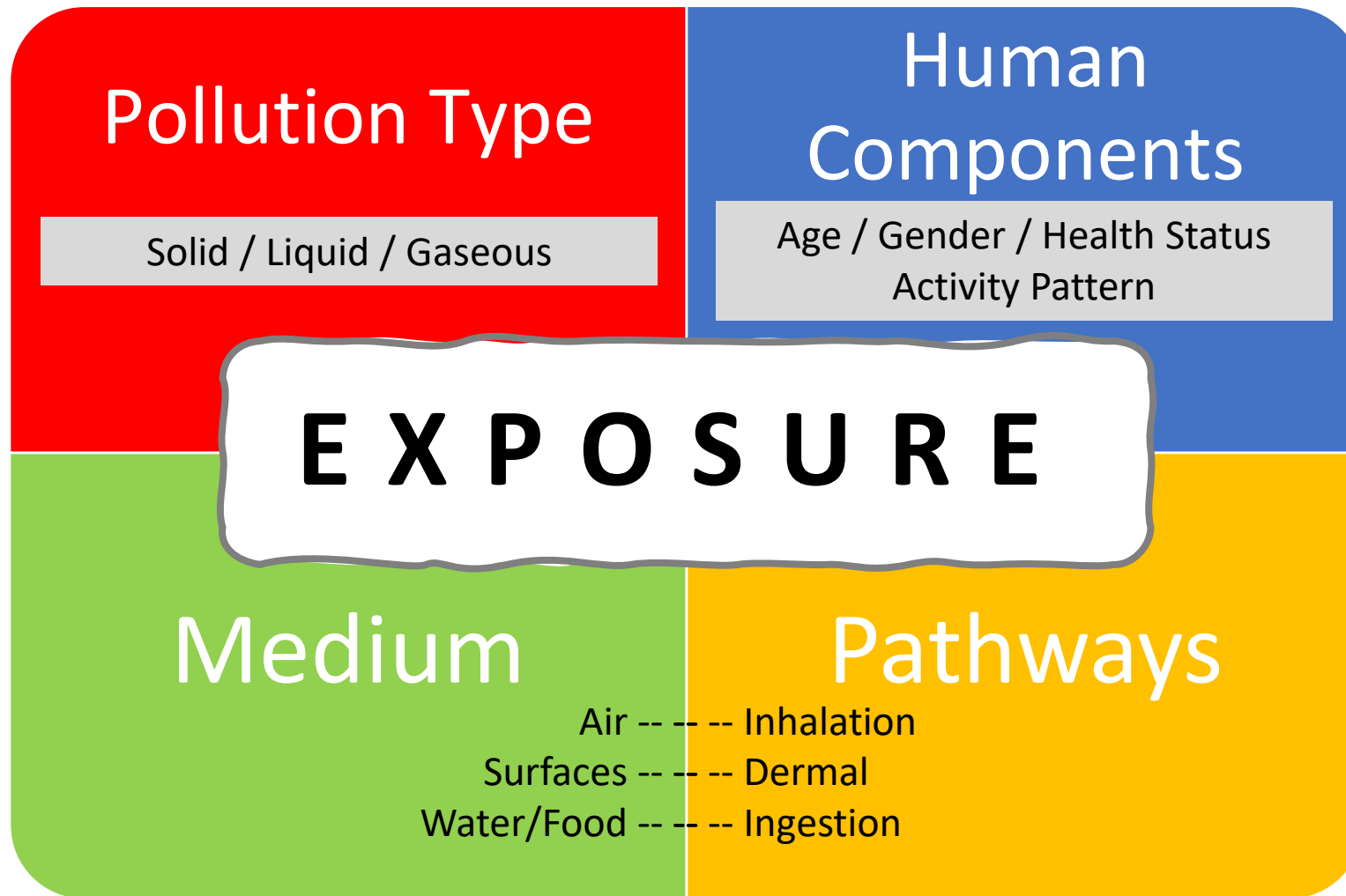
Location / Indoors

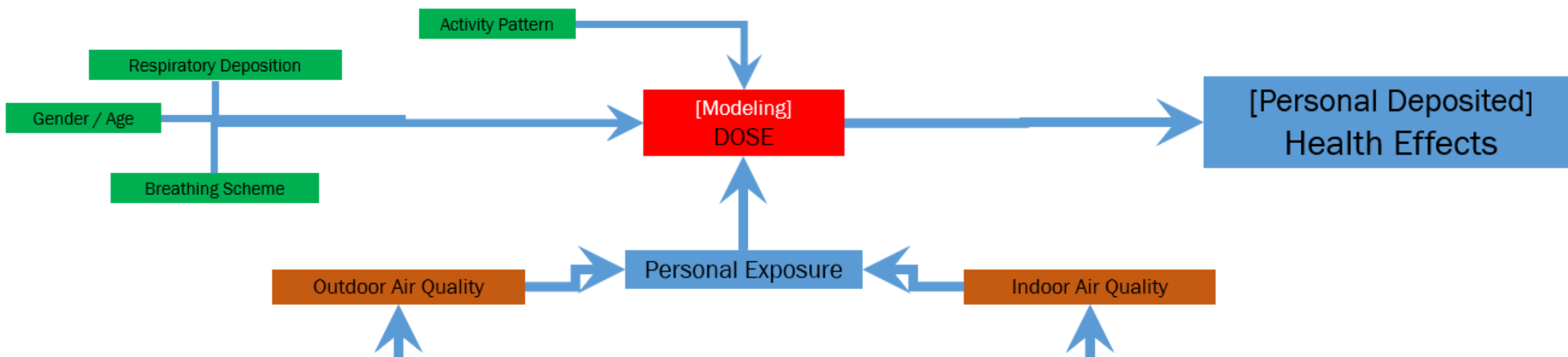


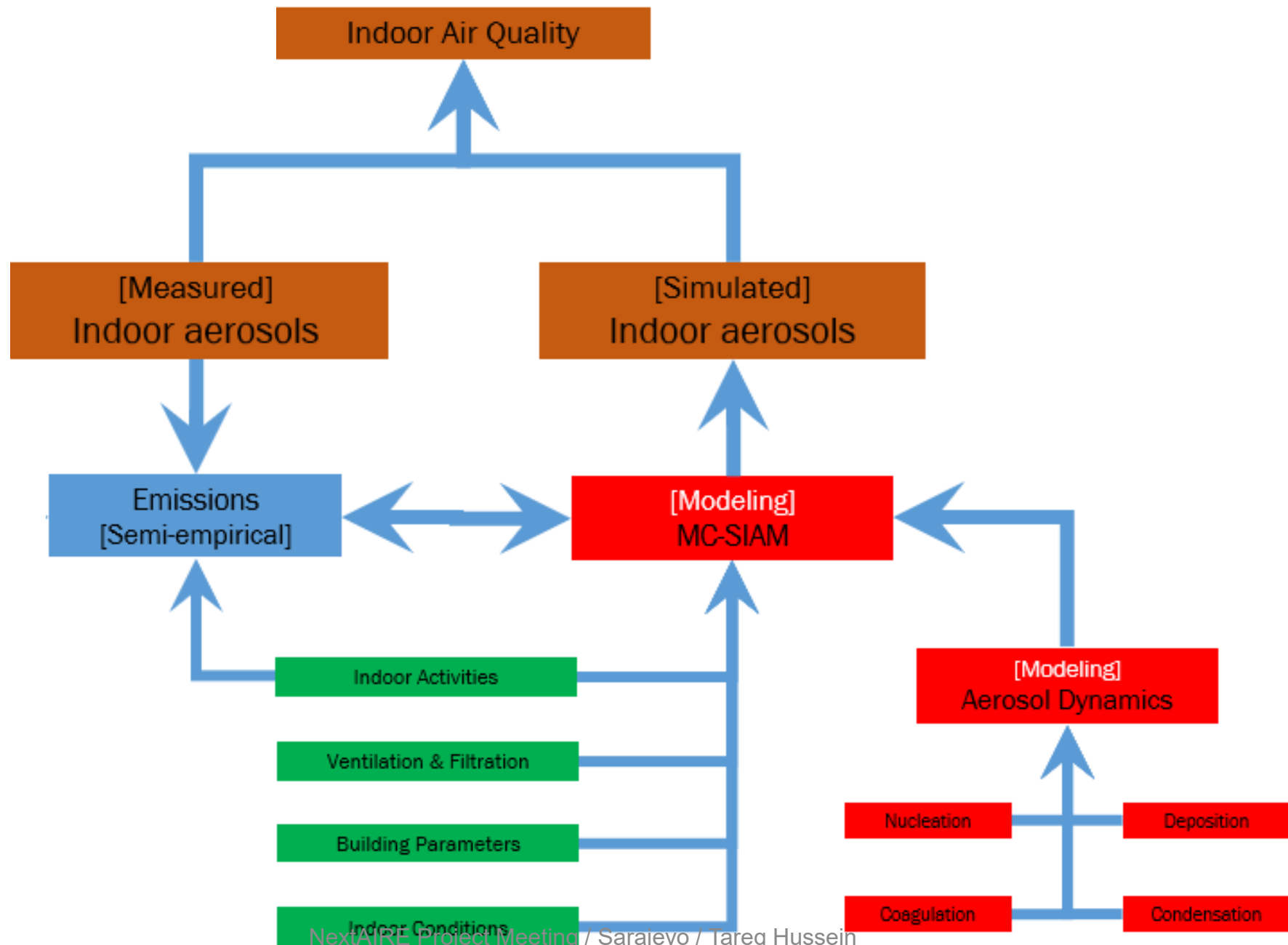
Location / Outdoors

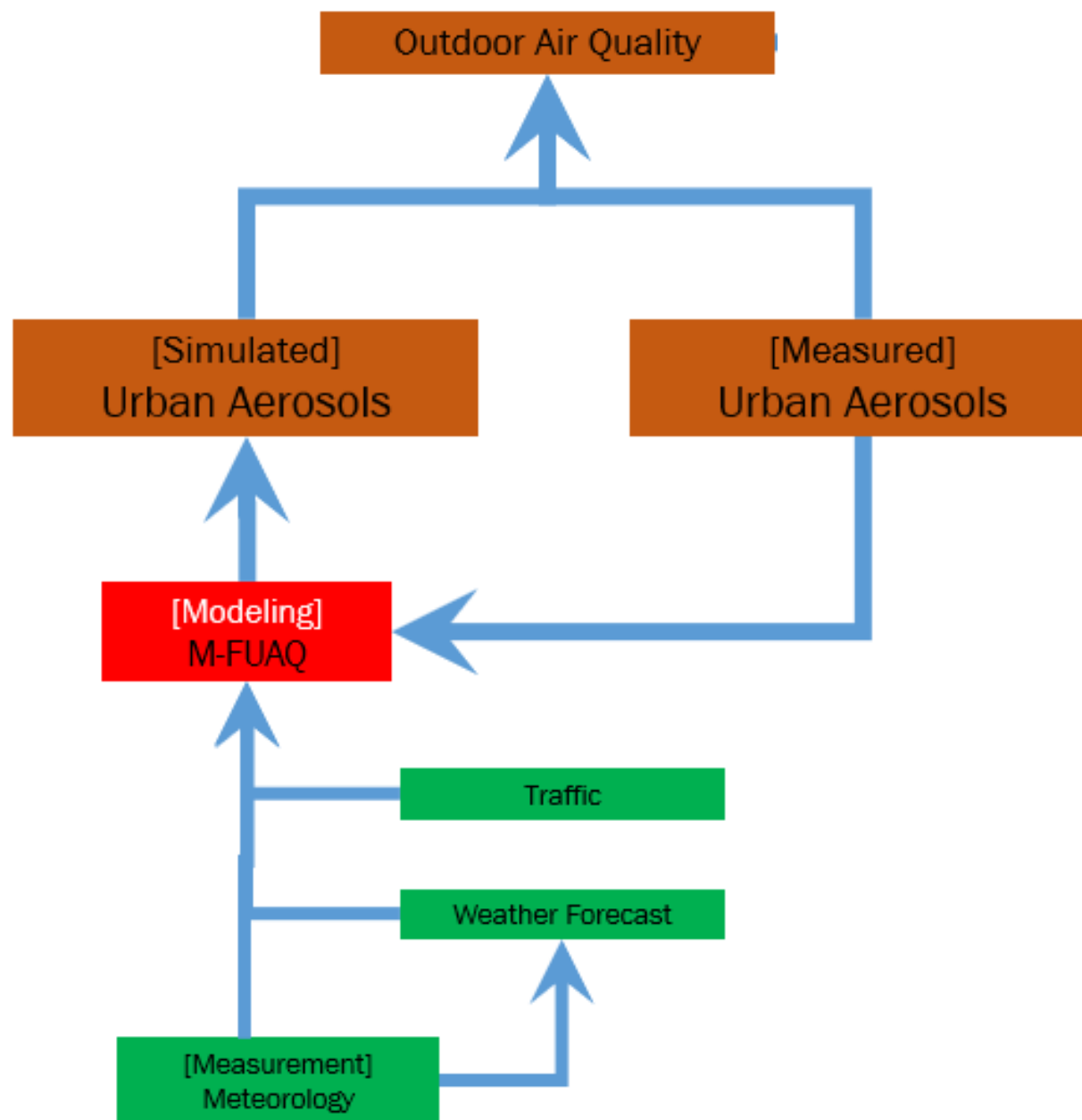


EXPOSURE OVERVIEW

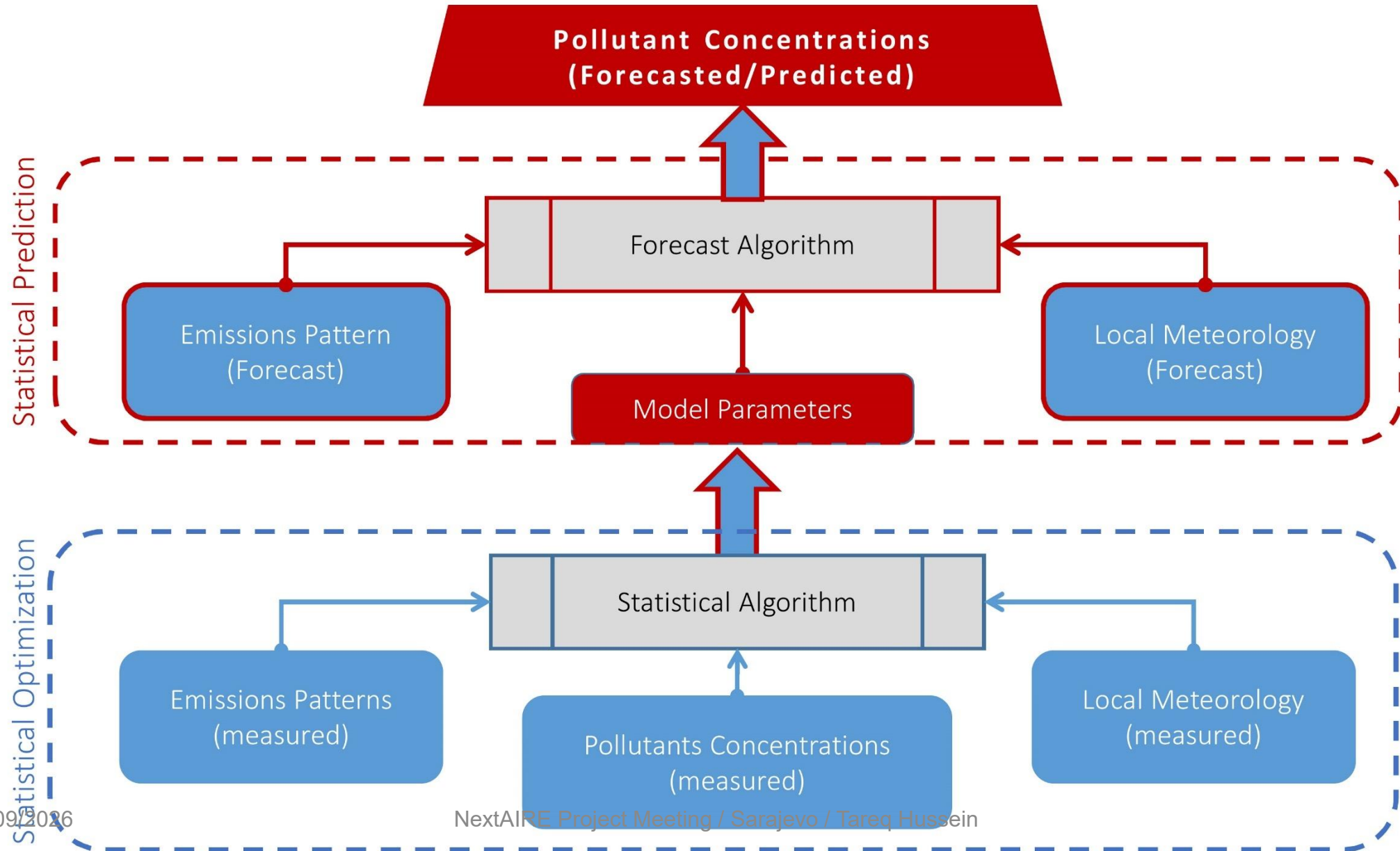




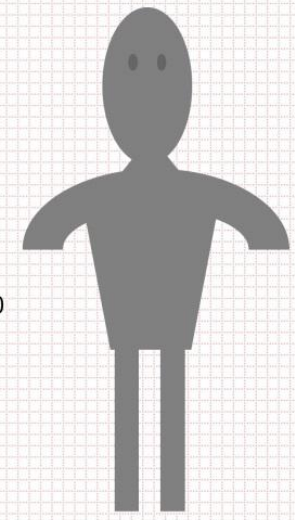
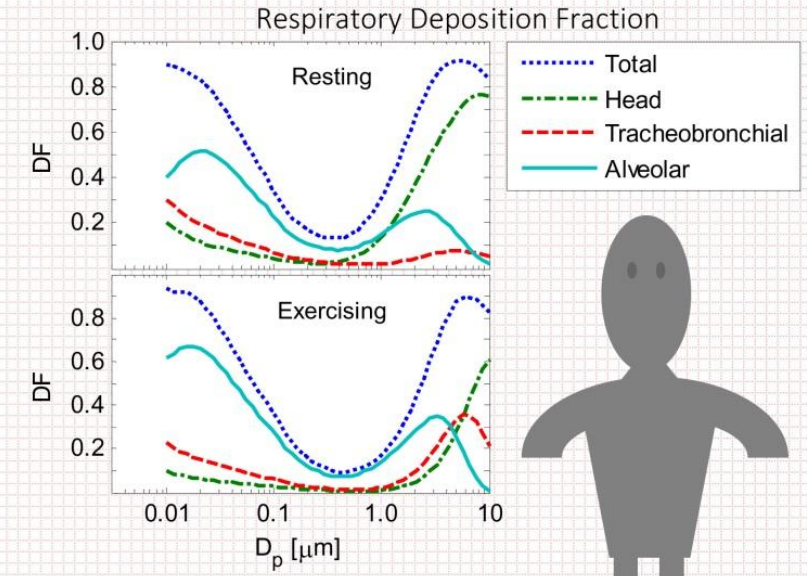
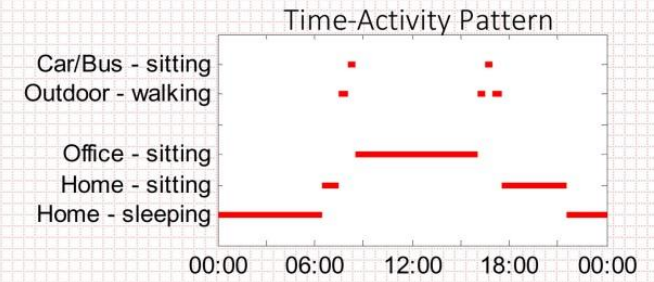
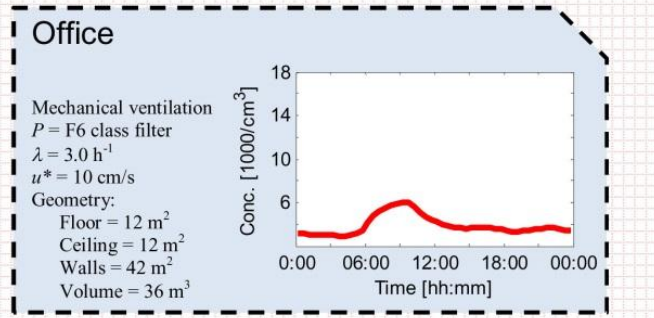
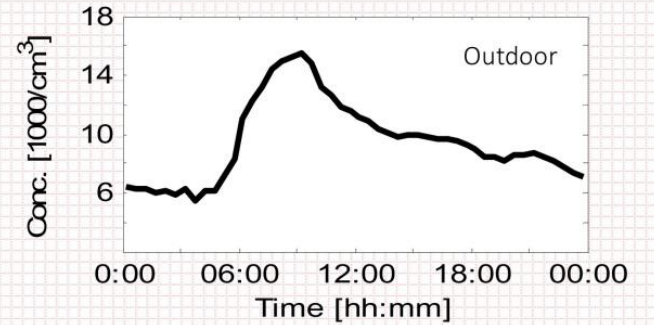
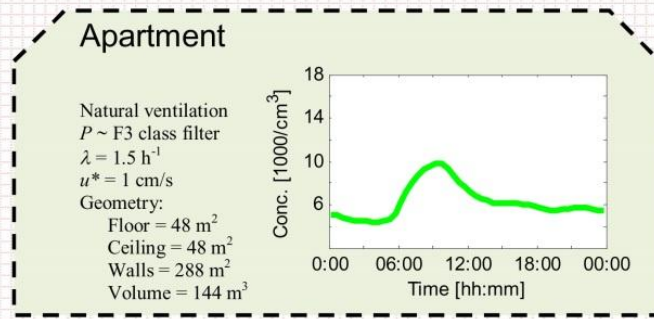




Urban Aerosol Forecast Model



Adult Male Urban Helsinki Workday



Minute Ventilation, V_E :
Walking = 23 L/min
Standing = 11 L/min
Sitting = 9 L/min

24-hours Deposited Dose in the Adult Male

	Particle Number [$\times 10^9$ particles]			Particle Mass [μg]		
	FP	UFP	ACCU	FP	UFP	ACCU
Total	59.51	56.20	3.40	42.29	1.57	40.77
Head	7.51	6.50	1.02	18.52	0.21	18.32
TB	11.94	11.55	0.41	4.02	0.31	3.72
P	40.07	38.15	1.97	19.75	1.05	18.73

Which exposure pathway to consider?!

The route of exposure is very important in an exposure event.

Inhalation is the most rapid route of uptake, followed by dermal contact and then by ingestion.

The health effects may vary significantly among the exposure routes and activities.

The phase of the pollutant (vapor or condensed) is an additional factor that influences health effects from inhalation.

Remarks...

- A low-cost sensor (LCS) can be deployed with large numbers in indoor and outdoor environments as well as on movable platforms
- LCS can be operated as a sub-network in parallel with other available reference air quality monitoring networks
- LCS can be calibrated in the field against a reference instrument/station or against a nearby calibrated LCS
- A Calibrated LCS monitoring capabilities can be boosted up to virtually monitor additional air quality indicators
- The data-base produced by LCS that are deployed in large numbers can be utilized in exposure assessment

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Thank You!